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1 Executive Summary

1.1 AI-Based Intrusion Detection Systems for Next-Generation Networks

PhD Research Proposal

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Date: December 22, 2025

Duration: 4 Years

1.2 Research Context and Significance

The cybersecurity landscape is undergoing a fundamental transformation driven by the proliferation of next-generation networks (NGN). Technologies such as 5G/6G, Internet of Things (IoT), Software-Defined Networks (SDN), Network Function Virtualization (NFV), and edge computing are revolutionizing how we communicate, compute, and connect. However, these advancements introduce unprecedented security challenges that traditional intrusion detection systems (IDS) are ill-equipped to address.

Current IDS solutions suffer from critical limitations: - **Signature-based systems** cannot detect zero-day attacks or polymorphic malware - **Anomaly-based systems** generate unacceptably high false positive rates (10-20%) - **Static models** fail to adapt to evolving threat landscapes and concept drift - **Centralized architectures** create bottlenecks and privacy concerns - **Black-box ML models** lack explainability, hindering security analyst trust and adoption - **Resource-intensive processing** struggles with gigabit-speed traffic and real-time requirements

This research addresses these critical gaps through the development of an advanced AI-based intrusion detection framework that combines cutting-edge deep learning, adaptive mechanisms, distributed architectures, and explainable AI to provide superior threat detection across diverse network environments.

1.3 Problem Statement

How can we develop an AI-based intrusion detection system that achieves high accuracy (>98%), low false positive rates (<1%), real-time performance (<100ms latency), scalability across heterogeneous networks, adaptability to evolving threats, and transparent decision-making while preserving data privacy?

This multifaceted challenge requires innovations in: 1. Deep learning architectures for multi-scale feature extraction 2. Adaptive learning mechanisms for continuous improvement 3. Distributed detection frameworks for edge computing 4. Explainable AI for analyst trust and regulatory compliance 5. Privacy-preserving techniques for federated threat intelligence

1.4 Research Objectives

1.4.1 Primary Objective

Design, develop, and validate an advanced AI-based intrusion detection system for next-generation networks that surpasses existing solutions in accuracy, performance, scalability, adaptability, and explainability.

1.4.2 Specific Objectives

1. Hybrid AI Model Development

- Create CNN-LSTM hybrid architecture for spatial-temporal feature extraction
- Implement Transformer networks for capturing long-range traffic dependencies
- Develop autoencoders for unsupervised anomaly detection
- Design ensemble methods combining multiple AI paradigms

2. Adaptive Learning Framework

- Integrate reinforcement learning (DQN, A3C, PPO) for dynamic threat response
- Implement online learning for continuous model adaptation
- Develop transfer learning for cross-domain attack detection

- Create meta-learning for rapid adaptation to new threats
- 3. **Distributed IDS Architecture**
 - Design hierarchical edge-fog-cloud detection framework
 - Implement federated learning for privacy-preserving collaboration
 - Develop model compression for resource-constrained edge devices
 - Create secure aggregation protocols for distributed intelligence
- 4. **Explainable AI Integration**
 - Integrate SHAP, LIME, and attention mechanisms for interpretability
 - Develop visualization dashboards for security analysts
 - Create explainability metrics for IDS evaluation
 - Design human-in-the-loop feedback systems
- 5. **Comprehensive Validation**
 - Evaluate on benchmark datasets (NSL-KDD, CICIDS2017, UNSW-NB15, IoT-23)
 - Deploy physical testbed with 5G/IoT/SDN infrastructure
 - Conduct extensive cross-environment experiments
 - Compare against state-of-the-art baselines

1.5 Proposed Methodology

1.5.1 System Architecture

The proposed framework consists of seven interconnected layers:

1. **Data Collection Layer:** Network sensors, SDN controllers, IoT gateways capture multi-source traffic
2. **Preprocessing Module:** Feature extraction, normalization, encoding, and class balancing
3. **AI Detection Engine:**
 - CNN-LSTM hybrid for spatial-temporal analysis
 - Transformer networks for sequence modeling
 - Autoencoders for anomaly detection
 - Ensemble methods (Random Forest, XGBoost)
 - Adaptive learning (RL, online learning)
4. **Explainable AI Module:** SHAP, LIME, attention visualization
5. **Decision & Response Layer:** Alert generation, automated response, human interface
6. **Distributed Architecture:** Edge-fog-cloud hierarchy with federated learning
7. **Privacy Mechanisms:** Differential privacy, secure aggregation

1.5.2 Key Innovations

Hybrid Deep Learning Architecture:

CNN Layers → Extract spatial features from traffic matrices

LSTM Layers → Model temporal dependencies in sequences

Transformer → Capture long-range dependencies via attention

Ensemble → Combine predictions for robustness

Adaptive Learning: - Reinforcement learning agents learn optimal defense policies
- Online learning enables continuous model updates with streaming data - Transfer learning enables knowledge sharing across network domains

Federated Detection: - Local models train on edge devices without data sharing - Secure aggregation combines model updates at fog layer - Global model distributed back to edge for improved detection

1.5.3 Experimental Design

Datasets: - NSL-KDD: Classic benchmark with balanced classes (126K training samples) - CICIDS2017: Modern attacks with realistic traffic (2.8M flows) - UNSW-NB15: Hybrid synthetic-real attacks (2.5M records) - IoT-23: IoT-specific botnet traffic (325M packets) - Custom 5G dataset: Collected from testbed deployment

Testbed: - 4× NVIDIA A100 GPUs for deep learning training - 5× OpenFlow SDN switches with ONOS controller - 10× Raspberry Pi 4 edge nodes for distributed deployment - 50× IoT devices (sensors, cameras, smart devices) - Network monitoring infrastructure (Grafana, Prometheus)

Evaluation Metrics: - Detection Performance: Accuracy (>98%), Precision (>97%), Recall (>98%), F1-Score (>97%), FPR (<1%), AUC-ROC (>0.99) - Efficiency: Latency (<100ms), Throughput (>10K pps), CPU (<50%), Memory (<2GB) - Scalability: Performance vs. network size (100 to 100K devices) - Adaptability: Zero-day detection rate, concept drift handling - Explainability: Feature importance, analyst evaluation scores

1.6 Expected Impact

1.6.1 Academic Contributions

1. **Novel Architectures:** CNN-LSTM-Transformer fusion with attention for IDS
2. **Theoretical Frameworks:** Mathematical models for distributed IDS optimization
3. **Adaptive Learning:** RL-based dynamic threat response mechanisms
4. **Explainability Methods:** XAI techniques tailored for network security
5. **Publications:** 5-10 papers in top venues (IEEE S&P, USENIX Security, CCS, NDSS, IEEE TIFS)

1.6.2 Practical Impact

1. **Deployable Framework:** Open-source AI-IDS platform with comprehensive documentation
2. **Industry Guidelines:** Best practices for production deployment
3. **Tool Ecosystem:** Automated pipeline for feature engineering, training, evaluation
4. **Benchmark Datasets:** Curated 5G/IoT attack scenarios for reproducible research
5. **Testbed Blueprint:** Documented setup for academic and industry replication

1.6.3 Societal Benefits

- **Critical Infrastructure Protection:** Secure power grids, healthcare, transportation
- **Enterprise Security:** Enhanced threat detection for businesses
- **IoT Ecosystem Safety:** Protect billions of connected devices
- **Privacy Preservation:** Federated learning respects data sovereignty
- **Economic Value:** Reduce cybercrime costs (projected \$10.5T annually by 2025)

1.7 Timeline Overview

1.7.1 Year 1: Foundation (Months 1-12)

- Comprehensive literature review (100+ papers)
- PhD coursework completion
- Baseline model implementation (CNN, LSTM, RF, SVM)
- Initial experiments on NSL-KDD and CICIDS2017
- **Milestone:** Comprehensive exam passed, first paper submitted

1.7.2 Year 2: Development (Months 13-24)

- Hybrid architecture implementation (CNN-LSTM-Transformer)
- Adaptive learning integration (RL, online learning)
- Distributed framework design (federated learning)
- Testbed deployment preparation
- **Milestone:** Testbed operational, second publication submitted

1.7.3 Year 3: Validation (Months 25-36)

- Extensive experiments on all datasets
- 5G/IoT-specific validation
- Performance optimization (<100ms latency)
- Large-scale scalability testing
- **Milestone:** Framework v1.0 released, 3+ publications accepted

1.7.4 Year 4: Completion (Months 37-48)

- Additional experiments addressing reviewer feedback
- User studies for explainability evaluation
- Thesis writing and revision
- Defense preparation and final submission
- **Milestone:** PhD degree awarded, published papers

1.8 Resource Requirements

Budget Summary: \$53,000 over 4 years

Category	Amount
Computational Resources (Cloud GPU credits)	\$15,000
Network Testbed Hardware (SDN, IoT, servers)	\$18,000
Conference Travel (4 major conferences)	\$10,000
Software Licenses & Tools	\$2,500
Publications (Open access fees)	\$3,500
Miscellaneous (books, materials)	\$4,000

Funding Sources: - University research assistantship - Department research grants
- External fellowships (NSF GRFP) - Industry partnerships (equipment donations) -
Cloud provider research credits (AWS, Google)

1.9 Keywords

Intrusion Detection Systems, Next-Generation Networks, Deep Learning, Artificial Intelligence, 5G/6G Security, IoT Security, Software-Defined Networks, Edge Computing, Explainable AI, Federated Learning, Cybersecurity, Adaptive Learning, Reinforcement Learning, Network Security, Privacy-Preserving Machine Learning

1.10 Conclusion

This PhD research proposes a comprehensive, innovative approach to intrusion detection for next-generation networks. By integrating hybrid deep learning architectures, adaptive learning mechanisms, distributed deployment strategies, and explainable AI, the proposed system addresses critical limitations of existing IDS solutions.

The research is timely, addressing urgent security challenges in 5G/6G, IoT, and edge computing. The methodology is rigorous, with extensive validation across multiple datasets and real testbed deployment. The expected outcomes include both theoretical advances (novel algorithms, mathematical frameworks) and practical contributions (open-source platform, deployment guidelines, tools).

Upon completion, this research will deliver a deployable AI-IDS framework that achieves superior detection performance while maintaining transparency, scalability, and privacy—advancing both academic knowledge and practical cybersecurity capabilities for our increasingly connected world.

For detailed information, refer to the complete proposal document: `proposal/full_proposal.md`

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